

RMO(INDIA) – 1997

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let P be an interior point of a triangle ABC and let BP and CP meet AC and AB in E and F respectively. If $[BPF] = 4$, $[BPC] = 8$, and $[CPE] = 13$, find $[AFPE]$, where $[\]$ denotes area.
 2. For each positive integer n , define $a_n = 20 + n^2$ and $d_n = GCD(a_n, a_{n+1})$. Find the set of all values that are taken by d_n and show by examples that each of these values is attained.
 3. Solve for real x : $\frac{1}{[x]} + \frac{1}{[2x]} = (x) + \frac{1}{3}$, where $[x] =$ is the greatest integer less than or equal to x and $(x) = x - [x]$.
 4. In a quadrilateral ABCD, it is given that AB is parallel to CD and the diagonals AC and BD are perpendicular to each other. Show that:
 - (a) $AD \cdot BC \geq AB \cdot CD$.
 - (b) $AD + BC \geq AB + CD$.
 5. Let x , y and z are three distinct real positive numbers. Determine with proof whether or not the three real numbers $\left| \frac{x}{y} - \frac{y}{x} \right|, \left| \frac{y}{z} - \frac{z}{y} \right|, \left| \frac{z}{x} - \frac{x}{z} \right|$ can be length of the sides of a triangle?
 6. Find the number of unordered pairs $\{A, B\}$ (i.e. the pairs $\{A, B\}$ and $\{B, A\}$ are considered to be same) of subsets of an n -element set X which satisfies the conditions:
 - (a) $A \neq B$.
 - (b) $A \cup B = X$.

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